MTH101 Premium Summary: The Exam Buster

Module 1: Sets, Indices & Logarithms (Questions 1-10)

Q1. Power Sets

Question: If Set A = {1, 2}, what is the Power Set of A, denoted as P(A)?

Explanation:

Many students forget that a Power Set must include the "Empty Set" and the set itself. The Power Set is simply the "Set of all subsets".

- **Formula:** If a set has n elements, the Power Set has 2^n elements.
- Here, n = 2 (elements are 1 and 2). So we expect 2^2 = 4 subsets.

Solution:

1. List all subsets:
   - The empty set: {}
   - Single elements: {1}, {2}
   - The full set: {1, 2}
2. Combine them into one big set.

Answer: P(A) = {{}, {1}, {2}, {1, 2}}

Q2. Venn Diagrams (Intersection)

Question: Given U = {1, 2, 3, 4, 5, 6, 7, 8, 9, 10}, A = {1, 2, 5, 7}, and B = {1, 3, 6, 7}. Find A n B (A intersection B).

Explanation:

The symbol "n" stands for Intersection. Think of a street intersection where two roads cross. We are looking for elements that are present in BOTH lists.

Solution:

1. Look at Set A: 1, 2, 5, 7
2. Look at Set B: 1, 3, 6, 7
3. Which numbers appear in both?

- Is 1 in both? Yes.
- Is 7 in both? Yes.

Answer: {1, 7}

Q3. Set Complements

Question: Using the same sets as above, find (A u B)' (The complement of A union B).

Explanation:

This question tests two things: Union (u) and Complement (').

- **Union (u):** Combine everybody together (no duplicates).
- **Complement (‘):** Everything in the Universe (U) that is **NOT** in the group.

Solution:

1. Find A u B first:

   Combine A and B: {1, 2, 3, 5, 6, 7}.

2. Find the Complement:

   Go through the Universe {1...10}. Cross out the numbers we just found in step 1.

- Remaining: 4, 8, 9, 10

Answer: {4, 8, 9, 10}

Q4. Indices (Multiplication)

Question: Simplify 32^(1/5) x 81^(1/4).

Explanation:

Don't use a calculator yet! Fractional indices mean Roots.

- x^(1/2) is Square Root.

- x^(1/5) is 5th Root (What number multiplied by itself 5 times gives x?).

Solution:

1. Simplify 32^(1/5):

   We know that 2 x 2 x 2 x 2 x 2 = 32. So, the 5th root of 32 is 2.

2. Simplify 81^(1/4):

   We know that 3 x 3 x 3 x 3 = 81. So, the 4th root of 81 is 3.

3. Multiply:

   2 x 3 = 6

   Answer: 6

Q5. Solving Exponential Equations

Question: Solve for x: 2^(x-1) = 16.

Explanation:

The Golden Rule of Indices: Make the bases the same. You cannot solve 2^x = 16 easily unless you realize that 16 is a member of the "2 family".

Solution:

1. Express 16 as a power of 2:

   16 = 2^4

2. Rewrite the equation:

   2^(x-1) = 2^4

3. Cancel the bases: Since the bases (2) are the same, the powers must be equal.

   x - 1 = 4

4. Solve for x:

   x = 4 + 1

Answer: x = 5

Q6. Logarithm Rules

Question: Evaluate log(25) + log(4). (Assume Base 10).

Explanation:

Never try to calculate log(25) by hand. Use the Addition Law:

log(A) + log(B) = log(A x B).

Addition becomes Multiplication!

Solution:

1. Combine the logs:

   log(25 x 4)

2. Calculate the inside:

   25 x 4 = 100

3. Evaluate log(100):

   Ask yourself: "10 raised to what power gives 100?" (10^x = 100).

   The answer is 2.

   Answer: 2

Q7. Solving Log Equations

Question: Find x if log_x(8) = 3.

Explanation:

This is about converting "Log Form" to "Index Form".

- Log Form: log_base(result) = power
- Index Form: base^power = result

Memorize this conversion! It saves lives.

Solution:

1. Convert to Index Form:

   x^3 = 8

2. Solve for x:

   Take the cube root of both sides.

   x = cube_root(8)

3. What number times itself 3 times gives 8? (2 x 2 x 2 = 8)

   Answer: x = 2

Q8. Surds (Rationalization)

Question: Simplify 2 / (sqrt(3) - 1) by rationalizing the denominator.

Explanation:

In Math, we hate having square roots at the bottom of a fraction. To fix it, we multiply the top and bottom by the Conjugate.

- If the bottom is sqrt(3) - 1, the conjugate is **sqrt(3) + 1**.

Solution:

1. Set up the multiplication:

   Multiply top and bottom by (sqrt(3) + 1).

2. Multiply the tops:

   2 * (sqrt(3) + 1) = 2sqrt(3) + 2

3. Multiply the bottoms (Difference of Two Squares):

   (sqrt(3) - 1)(sqrt(3) + 1)

   = (sqrt(3))^2 - (1)^2

= 3 - 1

= 2

4. Divide:

   (2(sqrt(3) + 1)) / 2

   Cancel the 2s.

   Answer: sqrt(3) + 1

Q9. Binary Operations

Question: An operation * is defined on real numbers as a * b = a + b - ab. Find 2 * 5.

Explanation:

Don't panic! The symbol * doesn't mean multiply here. It means "Follow the Rule". The question gives you a specific rule for this exam. Just substitute the numbers.

Solution:

1. Identify a and b:

   a = 2, b = 5

2. Plug into the formula (a + b - ab):

   = 2 + 5 - (2 x 5)

3. Calculate:

   = 7 - 10

   = -3

   Answer: -3

Q10. Sequence (AP)

Question: The 3rd term of an Arithmetic Progression (AP) is 12 and the common difference is 2. Find the First Term (a).

Explanation:

Remember the formula for the n-th term of an AP:

$T_n = a + (n-1)d$

- T_n is the value (12).
- n is the position (3rd).
- d is the difference (2).

Solution:

1. Substitute what we know:

   $12 = a + (3 - 1)(2)$

2. Simplify the brackets:

   $12 = a + (2)(2)$

   $12 = a + 4$

3. Solve for a:

   $a = 12 - 4$

   Answer: $a = 8$
   .

Module 2: Algebra, Polynomials & Quadratics (Questions 11-20)

Q11. Quadratic Factorization

Question: Solve the quadratic equation by factorization: $x^2 - 5x + 6 = 0$.

Explanation:

To factorize $ax^2 + bx + c$, we need two numbers that multiply to give c (6) and add to give b (-5).

- Possible pairs for 6: (1, 6) or (2, 3).

- To get -5, both must be negative: -2 and -3.

Solution:

1. Rewrite the middle term:

   $x^2 - 2x - 3x + 6 = 0$

2. Group terms:

   $x(x - 2) - 3(x - 2) = 0$

3. Factor out the bracket:

   $(x - 3)(x - 2) = 0$

4. Solve:

   $x - 3 = 0 \Rightarrow x = 3$

   $x - 2 = 0 \Rightarrow x = 2$

   Answer: $x = 2$ or $x = 3$

Q12. Sum and Product of Roots

Question: Without solving, find the sum of the roots (alpha + beta) of the equation $3x^2 + 6x - 9 = 0$.

Explanation:

For any quadratic equation $ax^2 + bx + c = 0$:

- **Sum of roots** = $-b / a$
- **Product of roots** = c / a

Solution:

1. Identify a, b, and c:

   $a = 3, b = 6, c = -9$.

2. Apply the formula for sum:

   Sum = $-6 / 3$

Sum = -2

Answer: -2

Q13. The Remainder Theorem

Question: Find the remainder when the polynomial $P(x) = x^3 + 2x^2 - 3x + 1$ is divided by $(x - 1)$.

Explanation:

The Remainder Theorem states that if you divide a polynomial $P(x)$ by $(x - a)$, the remainder is simply $P(a)$. You don't need to do long division. Just plug in the number.

Solution:

1. Identify a:

   Since divisor is $(x - 1)$, then $a = 1$.

2. Substitute $x = 1$ into the polynomial:

   $P(1) = (1)^3 + 2(1)^2 - 3(1) + 1$

   $P(1) = 1 + 2 - 3 + 1$

3. Calculate:

   $1 + 2 = 3$

   $3 - 3 = 0$

   $0 + 1 = 1$

   Answer: 1

Q14. The Factor Theorem

Question: Show that $(x + 2)$ is a factor of $P(x) = x^3 + 3x^2 + 3x + 2$.

Explanation:

If $(x + 2)$ is a factor, then plugging in $x = -2$ should give a result of zero.

Solution:

1. Substitute x = -2:

   P(-2) = (-2)^3 + 3(-2)^2 + 3(-2) + 2

2. Expand:

   (-8) + 3(4) + (-6) + 2

   -8 + 12 - 6 + 2

3. Calculate:

   -8 + 12 = 4

   4 - 6 = -2

   -2 + 2 = 0

4. Conclusion: Since the result is 0, it is a factor.

   Answer: Proven (Remainder is 0)

Q15. Change of Subject of Formula

Question: Make **r** the subject of the formula V = pi * r^2 * h.

Explanation:

This means we want "r" to stand alone on one side of the equals sign. We must move everything else (pi and h) to the other side by doing the opposite operations.

Solution:

1. Start with V = pi * r^2 * h.
2. Divide both sides by (pi * h) since they are multiplying r:

   V / (pi * h) = r^2

3. Remove the square power by taking the square root of both sides:

   r = sqrt(V / (pi * h))

   Answer: r = sqrt(V / (pi * h))

Q16. Simultaneous Equations (Linear)

Question: Solve for x and y:

$2x + y = 5$ (Eq. 1)

$x - y = 4$ (Eq. 2)

Explanation:

The easiest method here is Elimination because +y and -y will cancel each other out if we simply add the equations.

Solution:

1. Add Eq. 1 and Eq. 2:

   $(2x + x) + (y - y) = (5 + 4)$

   $3x = 9$

2. Solve for x:

   $x = 9 / 3 = 3.$

3. Substitute x=3 into Eq. 1 to find y:

   $2(3) + y = 5$

   $6 + y = 5$

   $y = 5 - 6$

   $y = -1$

   Answer: $x = 3, y = -1$

Q17. Linear Inequalities

Question: Solve the inequality: $2(x - 3) >= 10$.

Explanation:

Treat the ">=" sign exactly like an equals sign (=), unless you divide by a negative number (which flips the sign).

Solution:

1. Expand the bracket:

   2x - 6 >= 10

2. Move -6 to the other side (it becomes +6):

   2x >= 10 + 6

   2x >= 16

3. Divide by 2:

   x >= 8

   Answer: x >= 8

Q18. Quadratic Inequalities

Question: Solve x^2 - 9 < 0.

Explanation:

This asks: "Where is the curve x^2 - 9 below the x-axis?"

First, find the critical points where it equals zero (x^2 - 9 = 0).

Solution:

1. Solve x^2 - 9 = 0:

   x^2 = 9 => x = 3 or x = -3.

2. Test the regions:
   - Pick a number between -3 and 3 (e.g., 0): 0^2 - 9 = -9 (Negative - Correct).
   - Pick a number greater than 3 (e.g., 4): 4^2 - 9 = 7 (Positive - Wrong).
3. Since we want < 0 (negative), the answer is the region **between** -3 and 3.

   Answer: -3 < x < 3

Q19. Inverse Functions

Question: If $f(x) = 2x - 5$, find the inverse function $f^{-1}(x)$.

Explanation:

To find an inverse, replace $f(x)$ with y, then swap x and y, and finally solve for the new y.

Solution:

1. Let $y = 2x - 5$.
2. Swap letters:

   $x = 2y - 5$

3. Make y the subject:

   $x + 5 = 2y$

   $y = (x + 5) / 2$

   Answer: $f^{-1}(x) = (x + 5) / 2$

Q20. Partial Fractions

Question: Resolve $(5x - 1) / ((x - 1)(x + 3))$ into partial fractions.

Explanation:

We split the big fraction into two smaller, simpler fractions: $A/(x-1) + B/(x+3)$. We need to find the values of A and B.

Solution:

1. Set up the equation:

   $(5x - 1) / ((x - 1)(x + 3)) = A/(x - 1) + B/(x + 3)$

2. Clear denominators:

   $5x - 1 = A(x + 3) + B(x - 1)$

3. Find A: Let $x = 1$ (to eliminate B).

5(1) - 1 = A(1 + 3)

4 = 4A => A = 1

4. Find B: Let x = -3 (to eliminate A).

   5(-3) - 1 = B(-3 - 1)

   -15 - 1 = -4B

   -16 = -4B => B = 4

   Answer: 1/(x - 1) + 4/(x + 3)

Module 3: Trigonometry & Geometry (Questions 21-30)

Q21. Pythagoras Theorem

Question: A ladder 5m long leans against a wall. The foot of the ladder is 3m away from the base of the wall. How high up the wall does the ladder reach?

Explanation:

This forms a Right-Angled Triangle.

- Hypotenuse (Ladder) = 5m
- Base = 3m
- Height = ?

Solution:

1. Use the formula: **Hypotenuse^2 = Base^2 + Height^2**
2. Substitute the values:

   5^2 = 3^2 + h^2

3. Calculate squares:

   25 = 9 + h^2

4. Solve for h^2:

   h^2 = 25 - 9

   h^2 = 16

5. Find height:

   h = sqrt(16)

   h = 4

   Answer: 4m

Q22. SOH CAH TOA (Sine Rule)

Question: In a right-angled triangle, if the Hypotenuse is 10cm and the angle is 30 degrees, find the Opposite side.

Explanation:

Identify the sides relative to the angle 30 degrees:

- **Hypotenuse** = 10
- **Opposite** = ? (This is what we want)
- **Formula:** sin(theta) = Opposite / Hypotenuse (SOH)

Solution:

1. Set up the equation:

   sin(30) = x / 10

2. We know that sin(30) is 0.5.

   0.5 = x / 10

3. Cross multiply:

   x = 0.5 * 10

   x = 5

   Answer: 5cm

Q23. Trigonometric Identities

Question: Simplify the expression: $1 - \sin^2(x)$.

Explanation:

This comes from the "Golden Rule" of trigonometry:

$\sin^2(x) + \cos^2(x) = 1$

Solution:

1. Rearrange the rule to make $\cos^2(x)$ the subject:

   $\cos^2(x) = 1 - \sin^2(x)$

2. Therefore, the expression simplifies directly to $\cos^2(x)$.

   Answer: $\cos^2(x)$

Q24. Angles in a Polygon

Question: Calculate the sum of interior angles of a regular Hexagon (6 sides).

Explanation:

The formula for the sum of angles in any polygon with n sides is:

$\text{Sum} = (n - 2) * 180$

Solution:

1. Identify n (number of sides):

   Hexagon means $n = 6$.

2. Substitute into formula:

   $\text{Sum} = (6 - 2) * 180$

   $\text{Sum} = 4 * 180$

3. Calculate:

   Sum = 720 degrees

Answer: 720 degrees

Q25. Circle Geometry (Arc Length)

Question: Find the length of an arc of a circle with radius 7cm which subtends an angle of 90 degrees at the center. (Take pi = 22/7).

Explanation:

Think of an arc as a "slice of crust" from a pizza.

Formula: Length = (Angle / 360) * 2 * pi * r

Solution:

1. Substitute the values:

   Length = (90 / 360) * 2 * (22/7) * 7

2. Simplify fractions:

   90/360 reduces to 1/4.

   Length = (1/4) * 2 * 22 (since the 7s cancel out)

3. Calculate:

   Length = (1/4) * 44

   Length = 11

Answer: 11cm

Q26. Coordinate Geometry (Midpoint)

Question: Find the midpoint of the line connecting points A(2, 4) and B(6, 10).

Explanation:

The midpoint is just the average of the coordinates.

Formula: ((x1+x2)/2 , (y1+y2)/2)

Solution:

1. Average the x-values:

   (2 + 6) / 2 = 8 / 2 = 4

2. Average the y-values:

   (4 + 10) / 2 = 14 / 2 = 7

3. Combine them:

   (4, 7)

   Answer: (4, 7)

Q27. Coordinate Geometry (Gradient/Slope)

Question: Calculate the gradient (slope) of the line passing through (1, 2) and (3, 8).

Explanation:

Gradient (m) is "Change in Y" divided by "Change in X" (Rise over Run).

Formula: m = (y2 - y1) / (x2 - x1)

Solution:

1. Identify values:

   x1=1, y1=2

   x2=3, y2=8

2. Substitute:

   m = (8 - 2) / (3 - 1)

3. Calculate:

   m = 6 / 2

   m = 3

   Answer: 3

Q28. Equation of a Line

Question: Find the equation of a line with gradient 2 passing through the point (0, 3).

Explanation:

Use the slope-intercept form: $y = mx + c$

- **m** is the gradient (2).
- **c** is the y-intercept (where it crosses the Y-axis). Since the point is (0, 3), the intercept is exactly 3.

Solution:

1. Substitute $m = 2$ and $c = 3$ into the equation.

   $y = 2x + 3$

   Answer: $y = 2x + 3$

Q29. Perpendicular Lines

Question: If two lines are perpendicular, and the gradient of the first line is 3, what is the gradient of the second line?

Explanation:

If lines cross at 90 degrees (perpendicular), their gradients are negative reciprocals.

Rule: $m1 * m2 = -1$

Solution:

1. We know $m1 = 3$.
2. Flip it (1/3) and change the sign (-).

   $m2 = -1/3$

   Answer: -1/3

Q30. Distance Between Two Points

Question: Find the distance between points A(1, 1) and B(4, 5).

Explanation:

This is actually Pythagoras Theorem in disguise!

Formula: Distance = sqrt((x2 - x1)^2 + (y2 - y1)^2)

Solution:

1. Subtract the x's: 4 - 1 = 3
2. Subtract the y's: 5 - 1 = 4
3. Square them and add:

   3^2 + 4^2 = 9 + 16 = 25

4. Take the square root:

   sqrt(25) = 5

   Answer: 5

Module 4: Calculus (Questions 31-40)

Q31. Limits (Factorization Method)

Question: Evaluate the limit of (x^2 - 9) / (x - 3) as x approaches 3.

Explanation:

If you plug in x = 3 directly, you get 0/0, which is "undefined." We must simplify the algebra first by factorizing the top part.

Formula: Difference of Two Squares: a^2 - b^2 = (a - b)(a + b)

Solution:

1. Factorize the numerator x^2 - 9:

   (x - 3)(x + 3)

2. Rewrite the limit:

Limit = ((x - 3)(x + 3)) / (x - 3)

3. Cancel the (x - 3) terms:

   Limit = x + 3

4. Now plug in x = 3:

   3 + 3 = 6

   Answer: 6

Q32. Basic Differentiation (Power Rule)

Question: Find the derivative (dy/dx) of y = 3x^4.

Explanation:

Differentiation calculates the "Rate of Change."

Rule: Multiply by the old power, then reduce the power by 1.

If y = ax^n, then dy/dx = anx^(n-1).

Solution:

1. Multiply the coefficient (3) by the power (4):

   3 * 4 = 12

2. Reduce the power (4) by 1:

   4 - 1 = 3

3. Combine them:

   12x^3

   Answer: 12x^3

Q33. Differentiation of a Constant

Question: Find the derivative of y = 7x + 5.

Explanation:

- The derivative of "x" is just 1.
- The derivative of any "constant number" (like 5) is **0** (because a number doesn't change, so its rate of change is zero).

Solution:

1. Differentiate 7x:

   7 * 1 = 7

2. Differentiate 5:

   0

3. Add them up:

   7 + 0

   Answer: 7

Q34. The Chain Rule (Function inside a Function)

Question: Differentiate y = (2x + 1)^5.

Explanation:

Use the Chain Rule: "Differentiate the Outside x Differentiate the Inside".

- Outside: The power of 5.
- Inside: The (2x + 1).

Solution:

1. Outside: Bring down the 5 and reduce power to 4.

   5(2x + 1)^4

2. **Inside:** Differentiate (2x + 1). The answer is 2.
3. Multiply:

   2 * 5(2x + 1)^4

= 10(2x + 1)^4

Answer: 10(2x + 1)^4

Q35. Product Rule

Question: Find dy/dx if y = x^2 * sin(x).

Explanation:

When two things are multiplied (x^2 AND sin(x)), use the Product Rule:

Formula: dy/dx = u(dv/dx) + v(du/dx)

- Let u = x^2
- Let v = sin(x)

Solution:

1. Differentiate u: du/dx = 2x
2. Differentiate v: dv/dx = cos(x)
3. Substitute into formula:

   (x^2)(cos(x)) + (sin(x))(2x)

   Answer: x^2 cos(x) + 2x sin(x)

Q36. Basic Integration

Question: Find the integral of 3x^2 dx.

Explanation:

Integration is the reverse of differentiation.

Rule: Add 1 to the power, then divide by the new power.

Don't forget to add "+ C" (Constant of Integration).

Solution:

1. Increase power from 2 to 3:

x^3

2. Divide by the new power (3):

   $(3x^3) / 3$

3. Cancel the 3s:

   x^3

4. Add C:

   $x^3 + C$

   Answer: $x^3 + C$

Q37. Definite Integrals

Question: Evaluate the integral of 2x dx from limit 0 to 2.

Explanation:

"Definite" means we want a specific number answer, not a formula.

1. Integrate normally.
2. Plug in the top number.
3. Subtract the result of the bottom number.

Solution:

1. Integrate 2x:

   Power becomes 2, divide by 2.

   $(2x^2)/2 = x^2$

2. Plug in top limit (2):

   $2^2 = 4$

3. Plug in bottom limit (0):

   $0^2 = 0$

4. Subtract:

   4 - 0 = 4

   Answer: 4

Q38. Integration of Trig Functions

Question: Evaluate the integral of cos(x) dx.

Explanation:

You need to memorize the circle of differentiation/integration:

- Differentiate sin(x) -> cos(x)
- Therefore, Integrate cos(x) -> sin(x)

Solution:

The anti-derivative of cos(x) is sin(x).

Always add C for indefinite integrals.

Answer: sin(x) + C

Q39. Motion (Velocity from Displacement)

Question: A particle moves such that its distance s = t^3 - 2t. Find its velocity (v) at t = 3 seconds.

Explanation:

In physics/calculus:

- Velocity is the **derivative** of Distance (ds/dt).
- Acceleration is the derivative of Velocity (dv/dt).

Solution:

1. Find velocity function (differentiate s):

   v = 3t^2 - 2

2. Substitute t = 3:

   v = 3(3)^2 - 2

   v = 3(9) - 2

   v = 27 - 2

   v = 25

   Answer: 25 m/s

Q40. Area under a Curve

Question: Find the area under the curve y = 2x between x = 1 and x = 4.

Explanation:

"Area under the curve" just means "Calculate the Definite Integral".

We need to integrate 2x from 1 to 4.

Solution:

1. Integrate 2x:

   Result is x^2.

2. Top Limit (4):

   4^2 = 16

3. Bottom Limit (1):

   1^2 = 1

4. Subtract:

   16 - 1 = 15

   Answer: 15 square units

Module 5: Matrices & Statistics (Questions 41-50)

Q41. Matrix Addition

Question: Given Matrix A = [2, 3] and Matrix B = [4, 5]. Find A + B.

Explanation:

To add matrices, you simply add the numbers in the "matching positions."

- First number of A + First number of B.
- Second number of A + Second number of B.

Solution:

1. Add first elements: 2 + 4 = 6
2. Add second elements: 3 + 5 = 8

Answer: [6, 8]

Q42. Matrix Multiplication

Question: If Row Matrix A = [1, 2] and Column Matrix B = [3, 4] (vertical), calculate A x B.

Explanation:

Matrix multiplication follows the "Row by Column" rule.

Multiply the first element by the first, the second by the second, and then sum them up.

Solution:

1. Multiply (1 * 3) = 3
2. Multiply (2 * 4) = 8
3. Add them: 3 + 8 = 11

Answer: 11

Q43. Determinant of a 2x2 Matrix

Question: Find the determinant of the matrix M = [[2, 5], [1, 4]].

Explanation:

For a 2x2 matrix [[a, b], [c, d]], the determinant is calculated as:

Det = (a * d) - (b * c)

(Top-Left times Bottom-Right) MINUS (Top-Right times Bottom-Left).

Solution:

1. Identify values: a=2, b=5, c=1, d=4.
2. Multiply diagonals:

   (2 * 4) = 8

   (5 * 1) = 5

3. Subtract:

   8 - 5 = 3

   Answer: 3

Q44. Singular Matrices

Question: Find the value of k if the matrix [[k, 4], [3, 6]] is a singular matrix.

Explanation:

A "Singular Matrix" is a fancy way of saying "The Determinant is Zero".

So, set the determinant to 0 and solve for k.

Solution:

1. Calculate determinant:

   (k * 6) - (4 * 3) = 0

2. Simplify:

   6k - 12 = 0

3. Solve:

   6k = 12

   k = 2

   Answer: k = 2

Q45. Matrix Transpose

Question: Find the transpose of matrix A = [[1, 2], [3, 4]].

Explanation:

To "Transpose" a matrix, you swap the Rows and Columns.

- The first row becomes the first column.
- The second row becomes the second column.

Solution:

1. Row 1 is [1, 2]. Turn it vertical -> Column 1 becomes 1, 2.
2. Row 2 is [3, 4]. Turn it vertical -> Column 2 becomes 3, 4.
3. New Matrix: [[1, 3], [2, 4]]

Answer: [[1, 3], [2, 4]]

Q46. Calculating the Mean

Question: Find the mean (average) of the following numbers: 2, 4, 7, 8, 9.

Explanation:

Mean = (Sum of all numbers) / (Count of numbers).

Solution:

1. Sum: 2 + 4 + 7 + 8 + 9 = 30
2. Count: There are 5 numbers.
3. Divide: 30 / 5 = 6

Answer: 6

Q47. Calculating the Median

Question: Find the median of the set: 12, 5, 8, 3, 10.

Explanation:

The Median is the "Middle Number."

Crucial Step: You MUST arrange the numbers in order (Smallest to Biggest) before finding the middle.

Solution:

1. Sort the list: 3, 5, 8, 10, 12.
2. Find the middle:

   The number in the exact center is 8.

   Answer: 8

Q48. Calculating the Mode

Question: Find the mode of the distribution: 2, 5, 2, 7, 8, 2, 5.

Explanation:

The Mode is the number that appears most frequently.

Solution:

1. Count occurrences:
   - 5 appears twice.
   - 2 appears three times.
   - Others appear once.
2. Since 2 appears the most, it is the mode.

   Answer: 2

Q49. Probability (Dice)

Question: A fair six-sided die is rolled. What is the probability of getting a number greater than 4?

Explanation:

Probability = (Number of Successful Outcomes) / (Total Possible Outcomes).

Solution:

1. Total outcomes on a die: {1, 2, 3, 4, 5, 6} -> Total = 6.
2. Successful outcomes (Number > 4): {5, 6} -> Count = 2.
3. Calculate Probability:

   2 / 6

4. Simplify fraction:

   1 / 3

Answer: 1/3

Q50. Combinations (Selection)

Question: In how many ways can a committee of 2 students be selected from a group of 4 students?

Explanation:

When order doesn't matter (just selecting a group), use the Combination Formula:

nCr = n! / (r!(n - r)!)

- n = Total items (4)
- r = Items to choose (2)
- "!" means Factorial (e.g., 4! = 432*1).

Solution:

1. Substitute values: 4C2 = 4! / (2! * 2!)
2. Expand factorials:

   Top: 4 * 3 * 2 * 1 = 24

Bottom: (2 * 1) * (2 * 1) = 2 * 2 = 4

3. Divide:

   24 / 4 = 6

   Answer: 6 ways